



Machine Learning 2025/2026

Detecting breast cancer from blood samples

Intermediate Report

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1 Introduction

In this report we explain the implementation of a Machine Learning model capable of accurately identifying individuals diagnosed with breast cancer based on a blood sample dataset.

The dataset used in this study consists of data from 116 patients, including 64 diagnosed with breast cancer and 52 belonging to a healthy control group. The nine analyzed variables comprise parameters such as age, body mass index (BMI), glucose and insulin concentrations, the HOMA index, the levels of leptin, adiponectin, resistin, and MCP-1.

2 Methods and Methodologies

2.1 Data Partitioning

The data was initially divided into three subsets: a training set (70%), a test set (15%), and a validation set (15%). However, the test and validation sets were later merged, because during the first stage of the project there is no hyperparameter tuning, invalidating the need of a separate test and validation set. The classifiers were trained using the training set, while the test set is reserved for evaluating the classifiers' performance and generalization ability. This was done using a stratified split, ensuring that the proportion of classes (*Healthy* and *Cancer*) was maintained in both subsets.

After the partitioning, the data was normalized according to $\frac{x-\mu}{\sigma}$, so that each feature had a mean of 0 and a standard deviation of 1, ensuring comparability across variables with different value scales.

2.2 Feature Selection

Before applying the classifiers, it is necessary to evaluate the discriminative power of the original variables to identify those most relevant for distinguishing between the two classes. Firstly, we plot box/violing plots for each feature to see which are more discriminative. After, these were ranked and picked based on various feature selection methods

2.2.1 Kruskal-Wallis Test

The Kruskal-Wallis test is a non-parametric statistical test used to compare two or more independent samples. In this context, it was applied to assess the statistical relevance of each variable in distinguishing between the *Healthy* and *Cancer* classes. The null hypothesis (H_0) assumes that the distributions

of samples from different classes come from the same population, i.e., that the variable has no discriminative power. When the p-value is below 0.05, H_0 is rejected, concluding that the samples come from distinct populations and that the feature has discriminative ability.

The function `stats.kruskal` from the `scipy` library was used to compute the p-value for each feature. The features were ranked according to their score values, allowing the identification of those with the greatest discriminative power between healthy and cancer samples.

2.2.2 ROC Curve and AUC

The ROC (Receiver Operating Characteristic) method is particularly suitable for binary classification problems. The ROC curve represents the true positive rate (TPR) as a function of the false positive rate (FPR) for different decision thresholds. The area under the curve (AUC) provides a quantitative measure of the performance of the classifier.

An AUC value of 1 indicates perfect separation between classes, while a value of 0.5 indicates performance equivalent to random guessing. When the value is below 0.5 we simply switch the variables so that the inverse is the one that counts. Therefore the ROC/AUC analysis allows identifying variables with the highest discriminative capacity.

In this analysis, each clinical feature was individually evaluated by computing its ROC curve against the binary class labels (Healthy and Cancer). The `roc_curve` and `auc` functions from the `scikit-learn` library were used to calculate the False Positive Rate (FPR), True Positive Rate (TPR), and the corresponding AUC for each variable.

The results were plotted using the `Plotly` library, generating individual ROC curves for all features in a grid layout, with the AUC value annotated in each subplot. Finally, the features were ranked according to their AUC values, allowing the identification of those with the greatest discriminative power between healthy and cancer samples.

2.2.3 Correlation Matrix

The correlation matrix is used to assess linear relationships between variables. High correlations indicate redundancy, meaning that two features vary similarly. Analyzing this matrix helps select subsets of less correlated variables, reducing redundancy and improving classifier efficiency.

The correlation matrix of the nine features was computed to assess the degree of linear dependence between variables. This analysis was implemented using the `numpy.corrcoef` function and visualized with `Plotly Express` as a heatmap.

2.3 Feature Reduction

2.3.1 Principal Component Analysis (PCA)

PCA is an unsupervised dimensionality reduction technique that projects data onto a new set of orthogonal axes, called principal components (PCs), which maximize data variance. The goal is to find a reduced set of variables capturing most of the information contained in the original data with minimal variance loss. Eigenvectors define the direction (axis orientation), while eigenvalues represent the variance explained along those directions. Three criteria were used to determine the optimal number of components to retain:

- **95% Threshold:** select components that collectively explain at least 95% of the total variance;
- **Scree Criterion:** identify the point where the eigenvalue plot (scree plot) stabilizes, indicating components with marginal contribution;
- **Kaiser Criterion:** retain only components with eigenvalues greater than 1 (applicable to normalized data).

The PCA class from the `scikit-learn` library was then applied to compute the principal components, eigenvalues, and explained variance ratios.

For the 95% variance criterion, the cumulative explained variance was calculated, and the minimum number of components required to reach 95% of the total variance was identified. Using `Plotly`, a bar chart was created to display the variance explained by each principal component. Additionally a graph was created that shows all the samples projected onto the first principal component (PC1). The principal components were plotted against their explained variance, along with a threshold line at eigenvalue = 1 to apply the Kaiser criterion. Components with eigenvalues greater than one were retained, and the corresponding explained variance was computed. The Scree plot, generated using `Plotly`, was analyzed subjectively to determine the point at which the curve flattens out, indicating where additional principal components contribute little to the total variance.

2.3.2 Linear Discriminant Analysis (LDA)

LDA aims to project data into a lower-dimensional subspace that maximizes class separability while minimizing within-class variability. This supervised method finds a direction (or several) that maximizes the ratio between between-class variance and within-class variance, according to Fisher’s criterion. Since the number of discriminant directions that can be found is limited by the number of classes, LDA reduces the data dimensionality to at most (*number of classes* – 1). The result is a linear combination of the original variables, usable for both visualization and classification. The `LinearDiscriminantAnalysis` class from `scikit-learn` was then fitted to the dataset to compute the optimal discriminant direction that maximizes class separation. The data were projected onto the first linear discriminant (LDA1), and the resulting one-dimensional representation was visualized using a scatter plot, allowing a qualitative evaluation of the separation between healthy and cancer samples.

2.4 Classifiers

After feature selection and reduction, three distance-based classifiers were applied to the data:

2.4.1 Euclidean Distance Classifier

In this classifier, each unknown sample is assigned to the class whose mean is closest in terms of Euclidean distance, defined as:

$$||x - m||_e = \sqrt{\sum_{i=1}^d (x_i - m_i)^2}$$

where x represents the sample and m the class mean. This method assumes that all variables have equal weight and similar variance. In the implementation, the mean vector of each class was computed using the selected features, using `np.mean`. Each test sample were then classified by computing its Euclidean distance to both class means and assigning it to the closest class with `np.linalg.norm`. The classifier’s performance was evaluated by calculating standard metrics, including sensitivity, specificity, precision, F1-score and accuracy.

2.4.2 Mahalanobis Distance Classifier

The Mahalanobis distance is a more sophisticated metric that accounts for correlations among variables and differences in scale. It is defined as:

$$\|x - y\|_m = \sqrt{(\mathbf{x} - \mathbf{y})^T \mathbf{C}^{-1} (\mathbf{x} - \mathbf{y})}$$

where $\mathbf{C}$ is the covariance matrix of the variables. This classifier adapts to the true geometry of the data, being more robust when variables are correlated. In the implementation, the mean vector of each class was first computed using the selected features using `texttnp.mean`. The pooled covariance matrix of the features was then calculated with `texttnp.var` and inverted to account for correlations and differences in scale. Each test sample was classified by computing its Mahalanobis distance to the class means, taking into account the covariance structure, and assigning it to the closest class. Standard performance metrics, including sensitivity, specificity, precision, F1-score and accuracy, were subsequently calculated to evaluate the classifier.

2.4.3 Fisher LDA Classifier

The Fisher LDA classifier combines linear discriminant analysis with the Mahalanobis metric. It determines a weight vector $\mathbf{w}$ that maximizes class separation and defines a decision threshold. Decisions are based on the sign of the linear discriminant:

$$y = \mathbf{w}^T \mathbf{x} + b$$

where b is the bias term. This classifier is particularly effective for two-class problems and when distributions are approximately Gaussian. To implement the Fisher LDA classifier, several computational steps were followed. The mean vector of each class was then computed to represent its centroid in the multidimensional feature space. Subsequently, the within-class scatter matrices were calculated for both classes, quantifying the covariance of samples around their respective means. These matrices were summed to obtain the total within-class scatter matrix (S_w).

The optimal discriminant direction was determined by multiplying the inverse of this matrix with the difference between the class mean vectors, according to:

$$\mathbf{w} = S_w^{-1} (\mu_{Cancer} - \mu_{Healthy})$$

A bias term (b) was then computed to position the decision boundary approximately midway between the two class means, given by:

$$b = -\frac{1}{2} \mathbf{w}^T (\mu_{Cancer} + \mu_{Healthy})$$

Finally, the classifier’s performance was evaluated using standard metrics, including accuracy, sensitivity, specificity, precision, and F1-score, providing a quantitative assessment of its discriminative ability.

2.5 Experimental Setup

The *main* script implements the complete data analysis and classification workflow. The dataset partitions are loaded and perform the feature selection and reduction tasks on the training split. Finally, multiple classifiers are fitted using the training data (all features, top 5 ranked using kruskal-wallis and ROC-AUC, PCAs, and LDA). These were tested for accuracy with the test split. In the euclidean distance classifier this is done by computing the mean feature vectors of the Healthy and Cancer classes from the training data and assigning each test sample to the class whose mean is closest in Euclidean distance. The Mahalanobis classifier extends this by considering feature correlations through the covariance matrix computed from the training data, allowing it to capture the underlying data structure more accurately when classifying unseen testing samples. Similarly, Fisher’s Linear Discriminant classifier determines the optimal projection axis that maximizes class separability based solely on the training set, and then applies this learned transformation to the testing samples to evaluate how well the model generalizes to new data. Additionally to the sensitivity, specificity, precision, F1-score, and accuracy metrics used to evaluate each classifier’s performance on the test samples, a ROC-AUC score was calculated using the same methods employed in the feature extraction tests (True Positive Rate and False Positive Rate using `roc_curve` from `scipy`). This metric provides an aggregate measure of each classifier’s discriminative ability across all decision thresholds, offering a more comprehensive evaluation of the overall performance.

To ensure the robustness and reliability of the results, each experiment was executed five times using different random seeds. For every run, the dataset was divided into random training and testing subsets using a stratified split, ensuring that the class proportions (Healthy and Cancer) were preserved across all partitions. After completing the five runs, the performance metrics score were averaged to obtain stable and representative results, minimizing the influence of random initialization and data partitioning variability.

3 Result Analysis

The following section presents the analysis of the obtained results, focusing on the performance of the implemented classifiers and dimensionality reduction methods.

Figure 1 illustrates the distribution of the extracted features for both Healthy and Cancer samples using violin plots. This visualization provides a detailed view of the data dispersion and density for each feature, highlighting differences in their statistical distributions between the two classes. The width of each violin indicates the probability density of the data at different values, allowing a direct comparison of variability and central tendency across features.

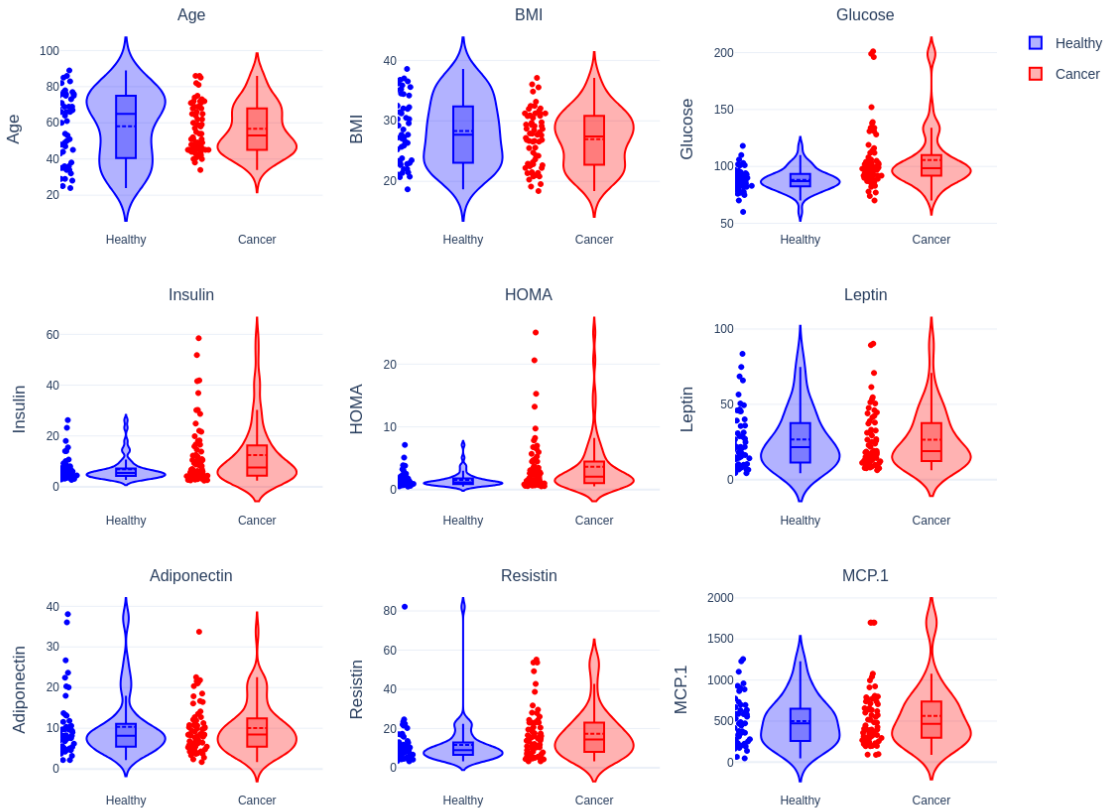


Figure 1: Violin plots.

The subplots clearly point towards some features being more discriminative

than others. Glucose, HOMA, resistin and Insulin have much higher differences in their value ranges than the others, more overlapping features. These observations are confirmed when doing the feature selection and dimensionality reduction.

To compare the performance of the classifiers, the performance metrics were first calculated using all features without any preprocessing. The results are presented in the table below.

Table 1: Classifier results using all the features.

Metric	Euclidean	Mahalanobis	Fisher LDA
Sensitivity (%)	50.5	65.3	63.2
Specificity (%)	80.0	82.5	82.5
Precision (%)	75.3	82.0	81.3
F1-score (%)	60.3	72.2	70.6
Accuracy (%)	64.0	73.1	72.0
ROC-AUC (%)	73.8	79.5	79.2

Analyzing Table 1, it is evident that some metrics, such as sensitivity, F1-score, and accuracy, fall below desired levels.

The ROC curves were plotted, and the AUC values were calculated for each feature, as shown in the figure below.

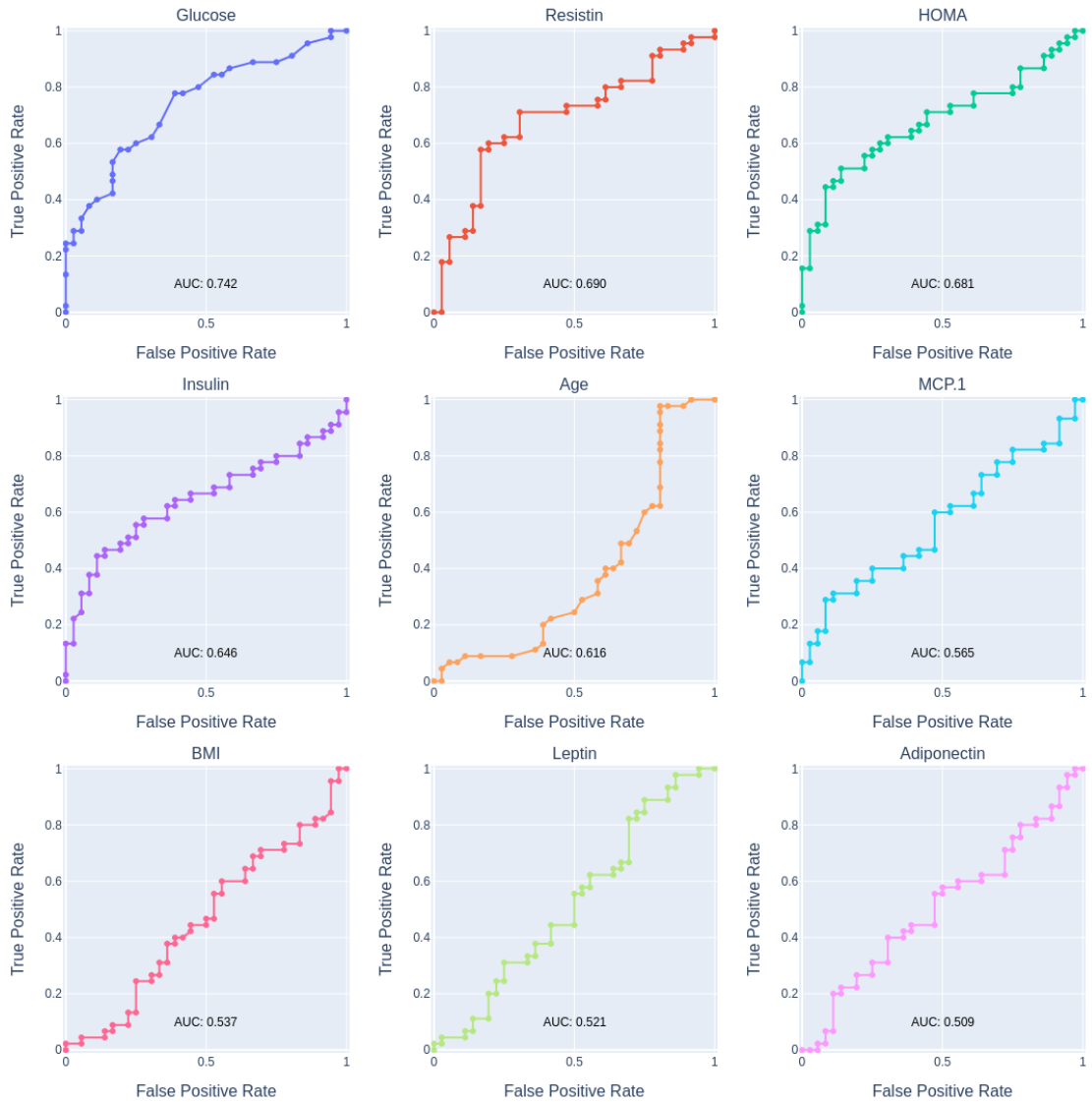


Figure 2: ROC curves for different feature sets.

According to Figure 2, the AUC values of all features were ranked in order to identify the top five features to be used in the training of the classifiers.

Table 2: Median ROC-AUC values across runs (sorted).

Feature	AUC
Glucose	0.7540
HOMA	0.6753
Resistin	0.6580
Insulin	0.6287
BMI	0.6148
Age	0.5438
MCP.1	0.5309
Leptin	0.5302
Adiponectin	0.5302

According to the Table 2, we can conclude that the top five features are Glucose, HOMA, Resistin, Insulin and BMI.

The different classifiers were applied using these features, and the resulting performance metrics for the test data are presented in the table below.

Table 3: Classifier results using the top 5 features (ROC selection).

Metric	Euclidean	Mahalanobis	Fisher LDA
Sensitivity (%)	52.6	63.2	62.1
Specificity (%)	81.2	75.0	75.0
Precision (%)	77.1	74.7	74.3
F1-score (%)	62.5	68.2	67.4
Accuracy (%)	65.7	68.6	68.0
ROC-AUC (%)	76.2	77.6	77.0

According to Table 3, the three classifiers yield similar values for each metric and occasionally show improvements compared to the results obtained using all features. However, this is not consistently observed.

We generated the correlation matrix to analyze how the features in training vary with respect to each other. This is to determine whether they are highly or weakly correlated and to assess potential redundancy among them.

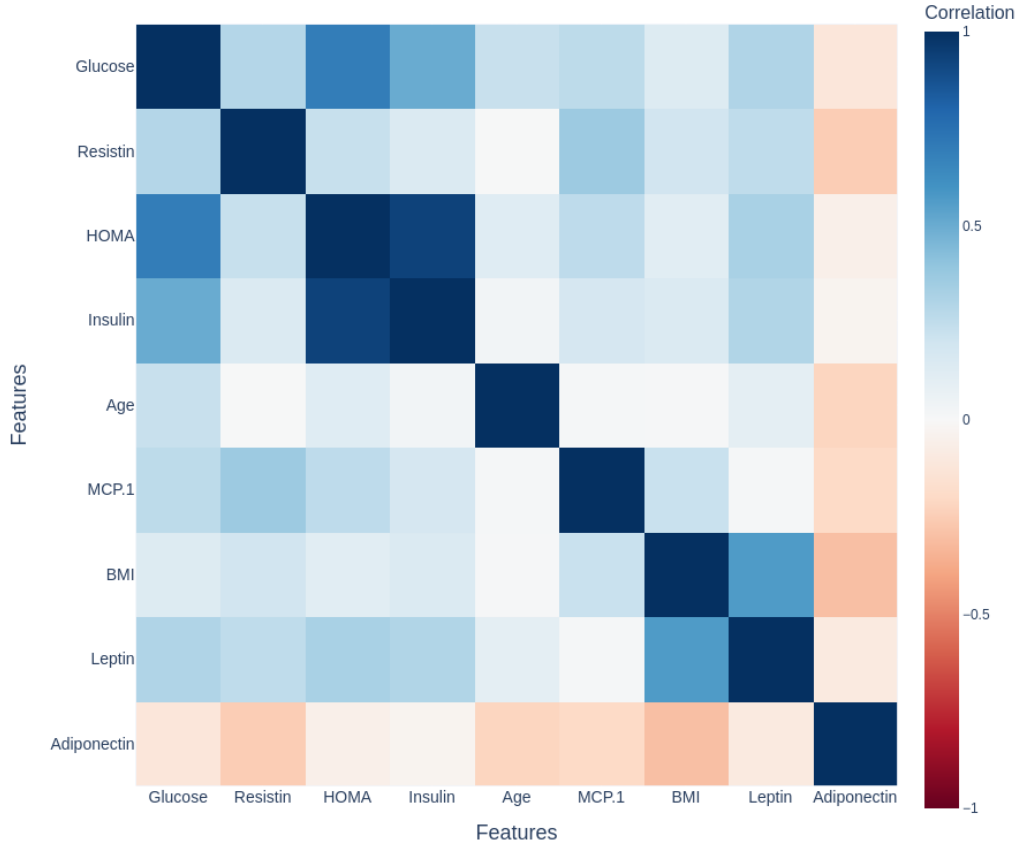


Figure 3: Correlation matrix between features.

From Figure 3, we can observe that Adiponectin is the most weakly correlated with the other features, followed by Age and BMI.

Next, we applied the Kruskal–Wallis test to rank the top five features by their H value. This indicates the degree to which each feature significantly differentiates between the groups, with higher H values reflecting stronger discriminatory power and greater relevance for distinguishing class differences.

Table 4: Median Kruskal-Wallis H-statistic values across runs (sorted).

Feature	H
Glucose	15.3190
HOMA	7.2860
Resistin	5.9201
Insulin	3.9271
BMI	3.1253
Age	0.4558
MCP.1	0.2259
Adiponectin	0.2169
Leptin	0.2169

From Table 4, we can identify the top five features as Glucose, HOMA, Resistin, Insulin and BMI, these were used on the classifiers.

The performance metrics on the test data for the classifiers applied to the feature selection using the Kruskal–Wallis test are presented in the table below.

Table 5: Classifier results using the top 5 features (Kruskal-Wallis selection).

Metric	Euclidean	Mahalanobis	Fisher LDA
Sensitivity (%)	52.6	63.2	62.1
Specificity (%)	81.2	75.0	75.0
Precision (%)	77.1	74.7	74.3
F1-score (%)	62.5	68.2	67.4
Accuracy (%)	65.7	68.6	68.0
ROC-AUC (%)	76.2	77.6	77.0

As can be seen in Table 5, the results are identical to those obtained using the ROC-AUC curves. This identical performance of the classifiers can be explained by the fact that both approaches identified the same five features. This demonstrates that, for this dataset, both feature selection methods successfully capture the features with the strongest class-discriminative power. To analyze the principal components, the three methods described in Section 2.3.1 were applied. The variance explained by each principal component is shown in the figure below.

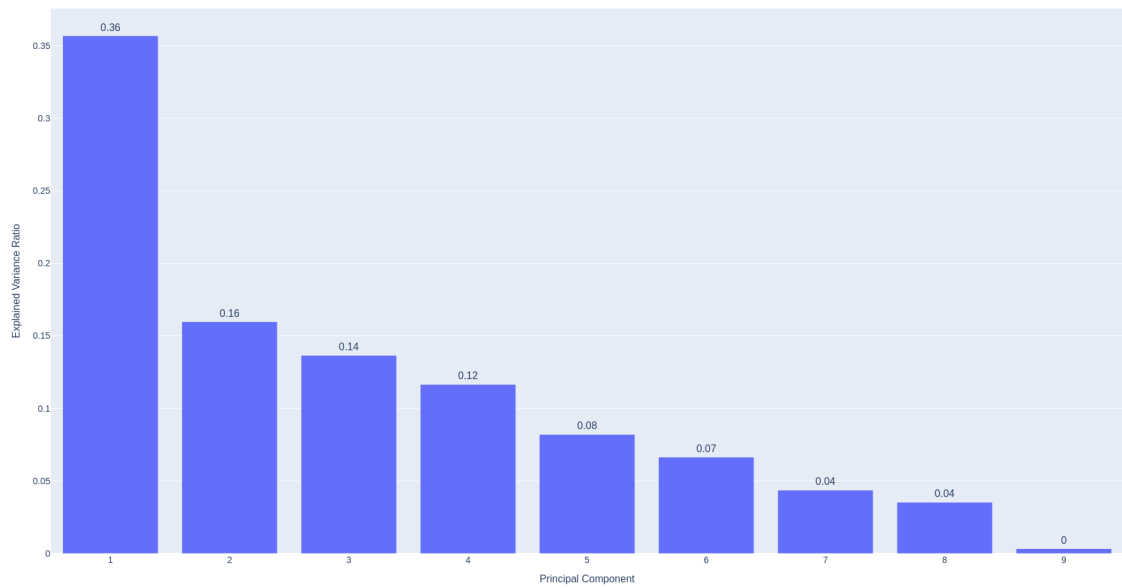


Figure 4: Explained variance for each principal component.

The 95% criterion indicated that the first seven principal components are required to retain most of the information and preserve the highest variance in the data. To visualize the selection of PCs according to the Kaiser criterion and the Scree test, the corresponding plot is shown below.

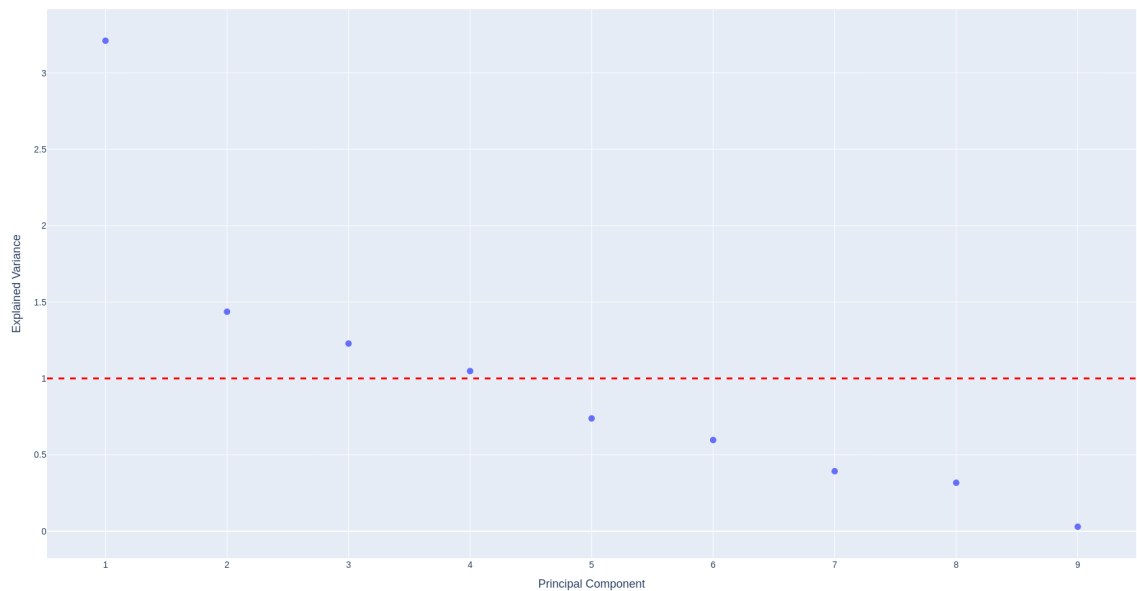


Figure 5: Principal components projected in a Kaiser and Scree plot.

By observing Figure 5, it can be seen that, according to the Kaiser criterion, only the first four principal components are selected. In contrast, the Scree method does not allow for a clear conclusion, as no leveling-off of the curve of the eigenvalues is observed. The first principal component, which accounts for the largest variance, is shown in the figure below.

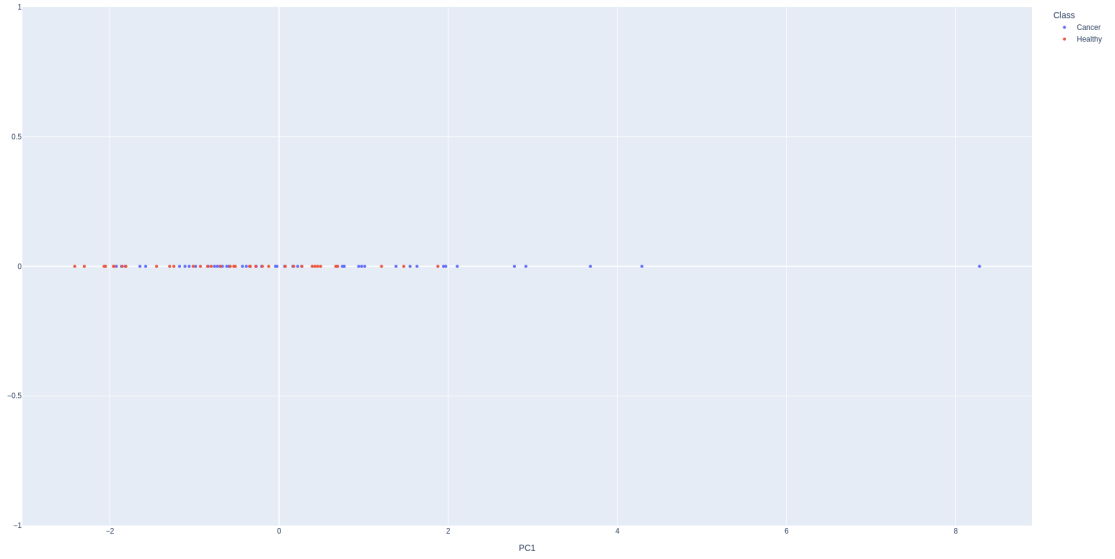


Figure 6: First principal component.

Among the three criteria used for PCA, the 95% variance threshold provided the best results, as it retained 7 components while preserving most of the information. The Scree criterion, on the other hand, suggested retaining only 4 components, which would likely result in a loss of information since the cumulative explained variance only reached a median of 78%, this is possibly due to not having a clear PC among the top 5 that explains most of the variance.

Because of this we applied the 7 components that explain 95% of the variance to our result analysis. Using the different classifiers on the testing data, the performance metrics were applied for this method, as presented in the table below.

Table 6: Classifier results using the top 7 principal components from PCA.

Metric	Euclidean	Mahalanobis	Fisher LDA
Sensitivity (%)	50.5	66.3	66.3
Specificity (%)	80.0	80.0	80.0
Precision (%)	76.0	80.2	80.2
F1-score (%)	60.3	72.2	72.2
Accuracy (%)	64.0	72.6	72.6
ROC-AUC (%)	73.6	75.9	75.7

According to Table 6, it can be observed that the results obtained with the Mahalanobis and Fisher LDA classifiers are identical, as explained in Section 3.1. Nevertheless, some metrics, such as accuracy, F1-score, sensitivity, precision, and specificity, show slight improvements compared to the results obtained using all features or the top features selected via ROC-AUC and Kruskal-Wallis. However, these improvements are only observed when using the Mahalanobis and Fisher LDA classifiers. With LDA, the data was reduced to a single dimension, as shown in the figure below.

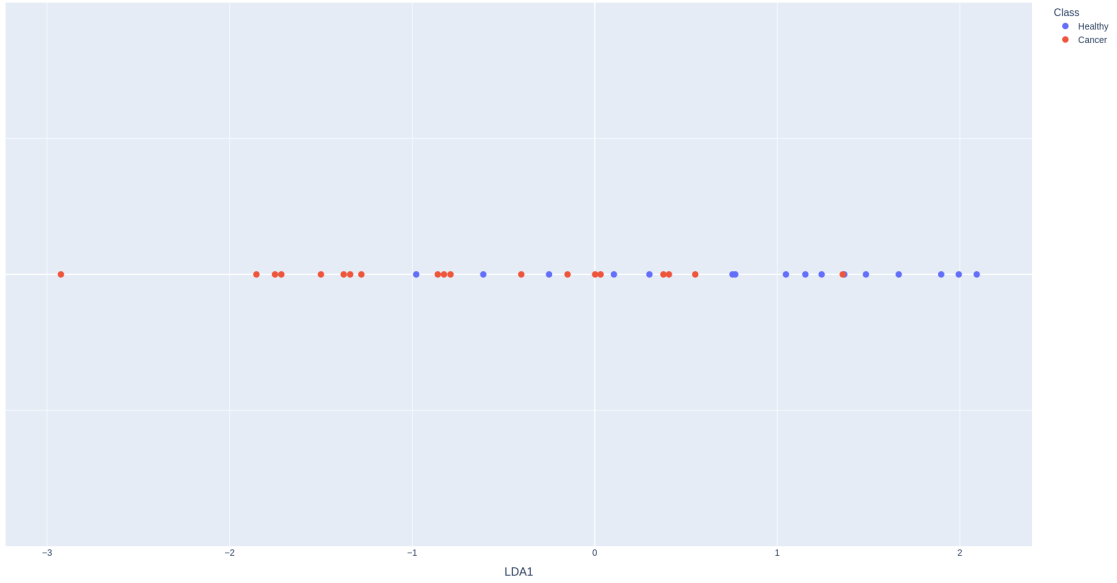


Figure 7: Linear Discriminant in one dimension.

By applying the classifiers with the testing data projected in the LDA, the resulting performance metrics are presented in the table below.

Table 7: Classifier results using the linear projection obtained by LDA (1D).

Metric	Euclidean	Mahalanobis
Sensitivity (%)	78.9	78.9
Specificity (%)	88.7	88.7
Precision (%)	89.4	89.4
F1-score (%)	83.7	83.7
Accuracy (%)	83.4	83.4
ROC-AUC (%)	90.9	91.8

Analyzing the values in Table 7, a clear improvement can be observed across all performance metrics for all classifiers, as expected using a supervised approach. It is also worth noting that the Fisher LDA classifier was not applied to the LDA features, as explained in Section 3.2.

3.1 Similarities between Mahalanobis and Fisher LDA classifiers

The identical results obtained with the Mahalanobis classifier and the Fisher LDA are theoretically expected, as both methods are mathematically equivalent under specific conditions. In the case of two-class classification problems, such as distinguishing Healthy from Cancer samples, and assuming that both classes follow approximately Gaussian distributions with equal covariance matrices, the Fisher LDA can be interpreted as a particular form of the Mahalanobis classifier. Both approaches rely on the same discriminant function, defined by a linear combination of features using the inverse of the within-class covariance matrix multiplied by the difference between the class mean vectors. Consequently, they define the same decision boundary in the feature space. In this dataset, these criteria are probably met, resulting in near identical results.

3.2 Fisher LDA not applied to LDA

Applying the Fisher LDA classifier after performing LDA-based feature reduction is unnecessary because the dimensionality reduction step already computes the optimal linear discriminant directions used by Fisher’s method. The transformed feature space produced by LDA inherently encodes the maximum class separability, making any subsequent Fisher classification redundant and mathematically equivalent to the projection already achieved.

4 Conclusion

The comparative analysis of feature selection and dimensionality reduction methods demonstrated that the classification performance depends on the quality and relevance of the extracted features. The violin plots and ROC-AUC analysis revealed that Glucose, HOMA, Resistin, and Insulin exhibit higher discriminative power between Healthy and Cancer groups. Both ROC-AUC ranking and the Kruskal–Wallis test consistently identified these same features as the most informative, confirming their statistical significance and relevance for class separation.

When all features were used, the classifiers achieved moderate performance, with sensitivity and F1-score values below the desired threshold. After feature selection, slight improvements were observed, although the enhancement was not substantial, suggesting that redundant or weakly correlated features had limited impact on model performance. PCA-based dimensionality reduction, especially when retaining seven components that explained 95% of the variance, led to a more compact representation while maintaining comparable classification results.

Among the classifiers, the Mahalanobis and Fisher LDA methods consistently produced identical results, as expected from their theoretical equivalence under equal covariance assumptions. Both outperformed the Euclidean classifier across all tests, highlighting the importance of incorporating covariance information in the classification process. The application of LDA for supervised dimensionality reduction proved particularly effective, as the resulting one-dimensional projection achieved the highest overall metrics, with clear improvement in accuracy, precision, and ROC-AUC.

Overall, the results indicate that supervised linear methods such as LDA, which exploit class label information, provide superior class separation compared to unsupervised approaches like PCA. These findings confirm that linear discriminant techniques can be effective for feature analysis, but not obtaining very good overall results.